

SKILLS AND LANGUAGES

- **Programming Language:** Hands-on Projects: Python, JavaScript; Coursework: Java, C, C++, SQL, Solidity
- **Techniques:** PyTorch, TensorFlow, MapReduce, Docker, Git, Tableau, MS Office
- **Spoken Language:** Mandarin (Native), English (Expert), Cantonese (Fluent), Hokkien (Conversant), Japanese (Beginners)

WORK EXPERIENCE

Quantitative Researcher **Alveria Capital, Singapore** **Apr 2025 -Present**

- Researched on intraday 1-minute-level long-only strategies for US stocks with high volatility, with live data feed
- Implemented stock selector in real-time market quotes, used Python to develop a automated back-test system that trades on multiple stocks in the same time with complete metrics, logs, and visualizations, at high performance

Quantitative Developer **Datalouder.com, Hong Kong** **June 2023 - Sept 2023**

- Used QuantLib to estimate delta , implement delta-neutral strategies based on US Stocks price and options data
- Developed mid-frequency trading Strategy based on Momentum Gap and Mean Reversion Strategies based on ETH/USDT HFT data, with additional techniques like DCA and periodical position reduction
- Performed passive market-making strategies on second-level ETH/USDT order fills data, set the latency and drift, and evaluate the performance with final PnL, inventory skew, and execution efficiency
- Implemented back-testing of strategies, built real-time data pipe for monitoring strategy performance and alerting on execution issues, with integration into the live web-based platform (Supports CI/CD), forwarded to Telegram channel
- Wrote automated Python script to fetch PB/PE ratios from Bloomberg, integrated to live database

Part-time Research Assistant **Nanyang Technological University** **Aug 2024 - Jan 2025**

- Constructed data warehouse and developed websites to visualize the global climate data using Tableau. Details, Website.

Full-time Research Staff **The Hong Kong Polytechnic University** **May 2024 - Sept 2024**

- Used LLMs to analyse reports about Firms' Mandatory Climate Risk Disclosures Effect on Mitigating Global Warming with empirical data by conducting Event Study with stock prices.Source Code

Summer Research Internship **The Chinese University of Hong Kong** **June 2022 - Dec 2022**

- Developed a universal benchmark for Deep Learning -Based Time Series Forecasting for different datasets. (Source Code).

Full-time Research Staff **Hong Kong Logistics Robotics Centre** **May 2021 – Sept 2021**

PROJECTS

- **Awesome Twitter Clone:** Built a full-stack twitter clone software on website with features about login/register/tweets/commenting/following, with React.js and MongoDB.
- **Refined FinRL:** Enhanced a reinforcement learning framework for multi-cryptocurrency trading at 1 hour level, optimizing strategy performance using the stable-baseline and gym library.
- **Backtest with Vis:** Built a trading backtesting framework in Python, with complete metrics and visualizations, posted to live websites using Poly.js

EDUCATION

Nanyang Technological University **Singapore** **Aug 2024 – Oct 2025 (Expected)**

- **M.Sc. in Financial Technology (Specialism: Intelligent Process Automation), CGPA:4.2/5.0**
- **Coursework:** Algorithmic Trading, Stochastic Modeling, Quantitative Methods in Finance, Statistical Modeling, Derivatives Pricing, Fixed Income Securities, Finance and Risk Management, Financial Markets, Reinforcement Learning, Natural Language Processing, Information Retrieval, Blockchain Engineering & Smart Contracts

The Chinese University of Hong Kong **Hong Kong** **Sept 2020 – Jul 2024**

- **B.Sc. (Hons) in Computer Science (Stream: Database and Information Systems), cGPA3.5/4.0, First Class (Top 15%)**
- **Coursework:** Design and Analysis of Algorithms, Fundamentals of AI, Web-scale Information Analytics, Database Systems, Cloud Computing & Storage, Computer Networks, Operating Systems, Software Engineering, Probabilities and Statistics

Honors

- **Postgraduate Entrance Scholarship 2024**, Anxi Mingning Educational & Charitable Association *Cap. 8,000 CNY.* (Details)
- **ELITE Scholarship AY 2021/2022**, Faculty of engineering, CUHK, *Cap. 20,000 HKD.* (Details)
- **Honors at Entrance AY2020/2021**, the Chinese University of Hong Kong, *Cap. 300,000 HKD.*(Details).
- **Undergraduate Entrance Scholarship 2020**, Anxi Mingning Educational & Charitable Association *Cap. 5,000 CNY.* (Details)